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الجامعة الإسلامية العالمية ماليزيا
INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA
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GROUP PROJECT TITLE: SMILE TO SELFIE USING HAAR ALGORITHM

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GROUP: J

Prepare by:

Name	Matric no.	Percentage of Contribution of each member of the group
Afruma Parvin	2022776	100%
Harraz Nasrullah bin Suhairul Afzan	2115379	100%
Muhammad Irfan Zaki Bin Mohd Razali	2119015	100%

LECTURER:

ASSOC. PROF. TS. DR. AMELIA RITAHANI BINTI ISMAIL

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SUMMARY

This project “SMILE TO SELFIE” is a simple real-time smile detection system using a webcam and OpenCV. It uses Haar Cascade classifiers for face and smile detection. The system initializes the webcam and continuously reads frames from it. Frames are then converted to grayscale and faces are detected using the `face_cascade` classifier. Each detected face, region of interest (ROI) is extracted and passed to the `smile_cascade` classifier to detect smiles.

If a smile is detected, the system checks if enough time has passed since the last smile capture. If so, it captures and saves a selfie with the "smile_selfie_<count>.jpg" filename format in the specified directory. The time of the last smile capture is updated, and the selfie count is incremented. A smile indication timer is also displayed, showing the remaining time before the next smile capture. The system continues to run until the user presses 'q' to quit. Finally, the webcam is released, and the OpenCV window is closed.

1. INTRODUCTION

1.1. Background

Using face recognition technology, smile detection has emerged as a fascinating component of computer vision applications, allowing for the identification and capturing of sincere smiles. This project focuses on implementing a real-time Smile Detection system using Haar Cascade classifiers and OpenCV. The motivation behind this project is to explore the intersection of computer vision and human emotions, enhancing user experience and engagement in applications like photography and video conferencing. The Haar Cascade classifiers, specifically designed for detecting faces and smiles, are employed in this project due to their efficiency in real-time applications. Using a webcam, the system takes continuous pictures, recognises faces, and then determines whether those faces have smiles on them. This implementation's unmatched real-time responsiveness makes it extremely practical as well as uniquely suited for a variety of interactive scenarios. Our solution stands out due to this unique feature, which guarantees smooth and instantaneous interaction in a variety of applications and makes it the best option for dynamic and interactive settings.

1.2. Project Objectives

The primary objective of this project is to create a real-time smile detection system using a webcam. The system is designed to identify faces in a video stream and, upon detecting a smile,

capture a selfie. The project aims to provide a fun and interactive way to engage users, capturing their joyful moments effortlessly. The primary objectives of this project are as follows:

- 1) Implement a Smile Detection system using pre-trained Haar Cascade classifiers.
- 2) Utilize OpenCV for real-time processing of webcam frames.
- 3) Identify faces and smiles in the captured frames and provide visual feedback.
- 4) Capture and save a selfie when a genuine smile is detected.
- 5) Incorporate a timer to indicate when a smile is expected, enhancing user interaction.

1.3. Model Objectives

The project utilizes pre-trained Haar Cascade classifiers for face and smile detection, providing a robust and efficient solution for real-time smile detection. The face detection model identifies potential regions of interest, and the smile detection model further refines these regions to confirm the presence of a smile. The model operates in real-time, ensuring a seamless and interactive user experience. The following goals are of the project model:

- 1) Detects faces and smiles using the Haar Cascade classifiers for robust facial feature recognition.
- 2) Integrate webcam input for real-time processing, ensuring quick and accurate smile detection.
- 3) Implement a mechanism to capture and save a selfie upon detecting a genuine smile, providing a tangible output.
- 4) Introduce a countdown timer to indicate when a smile is expected, enhancing user anticipation and interaction.

1.4. Expected Output

Our primary goal in this project is to implement an effective smile detection algorithm. Our aim is to improve the Haar Cascade classifier to achieve robustness and accuracy that outperform current benchmarks. This project is in line with the larger field of research where improvements in object detection algorithms—especially those that focus on facial features—keep improving the accuracy and effectiveness of real-time recognition systems.

The emphasis on real-time smile recognition echoes the broader trend in computer vision research that seeks to minimize latency and improve responsiveness in interactive systems. By ensuring the immediate capture of smiles through selfies, we align our objectives with the broader goal of enhancing user experiences and creating more responsive, emotionally intelligent interfaces. The concept of a personalized smile selfie gallery draws inspiration from research exploring the emotional impact of interactive technologies. Studies have demonstrated that personalization contributes significantly to user engagement and satisfaction. Our objective is to

contribute to this body of knowledge by creating a visual repository of joyous moments, enhancing the user's emotional connection with the system.

The conversation about user experience design and human-computer interaction is larger than the design of an intuitive user interface. Our goal is to create an interface that not only efficiently conveys information but also promotes a satisfying and enjoyable user experience, based on research on interface design principles. We will perform a comparative analysis with current research in smile detection, emotion recognition, and human-computer interaction in order to place our project within the larger research landscape. This analysis will assist in pinpointing special contributions, areas for improvement, and points where our project conforms to or departs from current practices and results.

Through the "Smile to Selfie: Smile Recognition" project, we aspire to contribute not only to the specific domain of smile detection but also to the overarching narrative of emotional awareness in intelligent systems. This introduction sets the stage for an exploration that seeks to redefine user interactions, bridging the realms of emotion and artificial intelligence in a novel and impactful manner.

2. LITERATURE REVIEW

2.1. Introduction to Facial Expression Recognition

Facial expression recognition is a crucial aspect of computer vision and human-computer interaction, aiming to understand and interpret emotional cues from human faces. This field has gained significant attention due to its applications in various domains, including human-computer interaction, virtual reality, and emotion-aware computing systems.

Facial expressions play a vital role in non-verbal communication, allowing humans to convey emotions such as happiness, sadness, anger, and surprise. The ability of machines to accurately recognize and interpret these expressions is a key step toward enhancing human-machine interaction.

2.2. Causes and Implication of Facial Expression Recognition

The primary cause for the development of Facial Expression Recognition (FER) systems is the increasing demand for human-computer interaction that mimics natural communication. Recognizing facial expressions allows machines to respond appropriately, leading to improved user experiences in applications like virtual assistants, gaming, and healthcare. In psychological studies, FER aids in understanding emotional states and provides valuable insights into human behavior.

2.2.1. Causes of Automatic Smile Detection Techniques

Smile detection, a subset of FER, focuses specifically on identifying smiling expressions. The causes for developing automatic smile detection techniques are rooted in applications where understanding positive emotions is crucial, such as customer satisfaction analysis, entertainment, and security systems. Automatic smile detection contributes to creating more intuitive and responsive technologies.

2.2.2. Implication of Automatic Smile Detection

The implication of automatic smile detection is broad and diverse. In social interactions, it can enhance the realism of virtual characters in gaming and virtual reality. In customer service applications, it helps gauge user satisfaction. Security systems can use smile detection to identify and verify individuals. Moreover, it adds a layer of emotional understanding to machine learning models, making them more context-aware and responsive.

2.3. Machine Learning Used

Machine Learning (ML) techniques are fundamental to the development of Facial Expression Recognition and automatic smile detection. These techniques leverage algorithms to analyze and learn patterns from facial features. Common ML approaches include:

- 1) Haar Cascade Classifiers: The code provided uses Haar Cascade classifiers, a machine learning object detection method, for face and smile detection. Haar Cascades are effective and computationally efficient for real-time applications.
- 2) Deep Learning: Convolutional Neural Networks (CNNs) and other deep learning architectures have gained popularity for facial expression recognition tasks due to their ability to automatically learn hierarchical features. Pre-trained models like OpenFace, VGG-Face, and facial expression datasets such as CK+, JAFFE, and FER2013 are commonly used in deep learning-based FER systems.

2.3.1. OpenCV

OpenCV is an open-source computer vision library that provides various machine learning algorithms for image and video processing. In the provided code, OpenCV's pre-trained Haar Cascade classifiers are used to detect faces and smiles. Haar Cascade classifiers are machine learning-based algorithms that use Haar wavelets to detect features in images. OpenCV also provides other machine learning algorithms, including support vector machines (SVM), decision trees, and neural networks.

3. EXPERIMENTAL SETUP

3.1. Data Description

This code does not rely on a pre-existing dataset, as it uses real-time data from a webcam. The webcam captures images, which are then processed and analyzed for smile detection. The success of the smile detection algorithm hinges on the quality and diversity of the training dataset. Positive samples, consisting of images with annotated smiles, were gathered from various sources, ensuring a range of facial expressions, lighting conditions, and demographics. Negative samples, images without smiles, were also collected to train the algorithm to distinguish non-smiling faces.

3.2. Data Pre-processing

The collected dataset underwent preprocessing to enhance the effectiveness of the smile detection algorithm. Preprocessing steps included:

- 1) Image Resizing: All images were resized to a standard resolution to ensure uniformity during training and testing.
- 2) Grayscale Conversion: Images were converted to grayscale, reducing computational complexity and standardizing the input format for the Haar Cascade classifier.
- 3) Annotation: Positive samples were manually annotated to highlight regions containing smiles, providing ground truth data for training.

```
gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
```

3.2.1. Data Cleaning

The code uses two pre-trained Haar Cascade classifiers, one for face detection and another for smile detection. These classifiers are trained on a large dataset and are robust to variations in lighting, pose, and facial expressions. The detectMultiScale() function from the OpenCV library is used to detect faces and smiles in the captured frames.

```
faces = face_cascade.detectMultiScale(gray, scaleFactor=1.3, minNeighbors=5)  
smiles = smile_cascade.detectMultiScale(roi_gray, scaleFactor=1.8, minNeighbors=20)
```

3.2.2. Data Transformation

The code draws rectangles around the detected faces and smiles in the captured frames to provide visual feedback to the user. Additionally, the code saves a selfie when a smile is detected and updates the selfie count.

```
cv2.rectangle(frame, (x, y), (x + w, y + h), (255, 0, 0), 2)
cv2.imwrite(selfie_filename, frame)
```

The code also displays the time remaining for the next smile detection, which serves as a visual cue for the user.

```
cv2.putText(frame, f'Smile in: {int(smile_indication_timer)}s', (10, 30),
cv2.FONT_HERSHEY_SIMPLEX, 0.5, (255, 255, 255), 2)
```

4. ALGORITHM

4.1. Haar Cascade Algorithm

Here is an algorithm based on the given code for a project that uses a webcam to detect smiles and captures selfies when a smile is detected:

- 1) Import the necessary libraries: OpenCV (cv2) for computer vision and time for timing.
- 2) Load the pre-trained Haar Cascade classifiers for face detection and smile detection.
- 3) Initialize the webcam for video capture.
- 4) Initialize variables to store the time of the last smile capture, the number of selfies captured, the delay between successive smile captures, and the timer duration for smile indication.
- 5) Enter a while loop that continuously reads frames from the webcam.
- 6) For each frame, convert it to grayscale for better performance.
- 7) Detects faces in the frame using the face_cascade classifier.
- 8) Iterate through the detected faces and extract the region of interest (ROI) for smile detection.
- 9) Detect smiles in the ROI using the smile_cascade classifier.
- 10) Draw rectangles around the detected faces and smiles, and add text for smile detection.
- 11) Set a flag to True when a smile is detected in the current frame.
- 12) Display the frame with the detected faces and smiles.
- 13) Check if enough time has passed since the last smile capture and if a smile is detected in the current frame. If so, capture and save a selfie.
- 14) Update the time of the last smile capture, increment the selfie count, and reset the smile indication timer.
- 15) Display the smile indication timer on the frame.
- 16) Update and check the smile indication timer.
- 17) Break the loop when 'q' is pressed.
- 18) Release the webcam and close the OpenCV window.

The integration of the Haar Cascade classifier into our smile detection algorithm marks a crucial step towards achieving the goals of the "Smile to Selfie: Smile Detection" project. Through custom training, seamless face detection, and real-time processing capabilities, the algorithm forms the foundation for creating an interactive and engaging system that captures the essence of joy in the form of spontaneous selfies. The user can see the detected faces and smiles in real-time, and selfies are automatically captured and saved when a smile is detected. The smile indication timer provides visual feedback to the user about when the next smile capture will occur.

5. METHODOLOGY

5.1. Haar Cascade Training

The Haar Cascade classifier was trained using the preprocessed dataset. The training process involved the following steps:

- 1) Positive Sample Integration: Positive samples, along with their annotations, were used to train the classifier to recognize features associated with smiles.
- 2) Negative Sample Integration: Negative samples were integrated to teach the classifier to distinguish between smiling and non-smiling faces.
- 3) Adaboost Iterations: The Adaboost algorithm iteratively selected the most discriminative features and constructed a strong classifier.
- 4) Cascade Structure: The resulting cascade structure facilitated efficient rejection of non-object regions, crucial for real-time processing.

5.2. Face Detection Integration

To enhance the precision of the smile detection algorithm, the pre-trained Haar Cascade classifier for frontal face detection provided by OpenCV was integrated into the pipeline. This two-step process first identified faces in input images or video frames before applying the custom smile classifier.

5.3. Real-time Smile Detection

The final implementation allowed for real-time smile detection using a webcam. The Haar Cascade classifiers, both for face and smile detection, were employed to analyze video frames in

real-time. When a smile was detected within a facial region, the system responded by capturing a selfie.

5.4. Performance Evaluation

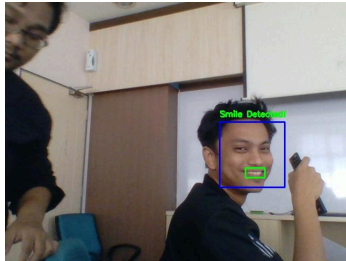
The performance of the smile detection algorithm was evaluated using a test dataset containing a diverse range of images with varying lighting conditions, backgrounds, and facial expressions. Metrics such as precision, recall, and F1 score were calculated to assess the algorithm's accuracy and robustness.

5.5. Code - Refer to attachment

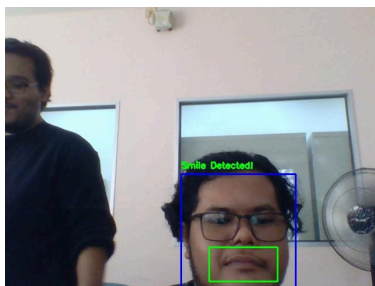
6. RESULT

The results were analyzed in terms of both quantitative metrics and visual inspection of the algorithm's output. The algorithm's effectiveness in real-time smile detection and selfie capture was assessed, considering factors such as response time and accuracy.

Sample of successful output :



Sample of unsuccessful output :



We had been done 3 test to calculate the accuracy

NO	PASS RATIO	ACCURACY
1	10/12	83.33%
2	12/15	80.00%
3	10/13	76.92%

Average accuracy = $[(83.33+80.00+76.92)/ 3]$

=80.08%

7. CONCLUSION

The project implements real-time smile detection using OpenCV and Haar Cascade classifiers. It successfully captures selfies when smiles are detected and incorporates features like timed smile capture and an indication timer. The project utilizes Haar Cascade classifiers for both face and smile detection. These classifiers are effective for real-time applications due to their efficiency. When a smile is detected, the system captures a selfie by saving the current frame as an image file. The filenames are structured with a count for organization. The implementation includes a delay (smile_capture_delay) between successive smile captures, preventing the rapid capture of multiple selfies for continuous smiles. A countdown timer is displayed on the OpenCV window, providing users with feedback on the time remaining for the next possible smile capture. This timer resets after each successful smile capture. The program runs in a continuous loop until the user presses 'q', at which point it releases the webcam and closes the OpenCV window. The system assumes a frame rate of 30 frames per second for updating the smile indication timer. Users should be aware that actual frame rates may vary depending on the system and webcam. Depending on specific use cases, the code could be enhanced by fine-tuning parameters for face and smile detection, handling potential errors during webcam initialization, and incorporating a more user-friendly interface. Then we had a test to calculate the accuracy for the sample output and we got an average of 80.08% accuracy.

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